

Structure and Properties of Organic Polymers

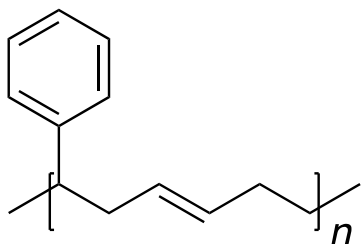
Organic Chemistry II

CHEM 2312

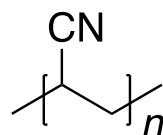
Dr. Michael Evans

27.1 Organic Polymers are Everywhere

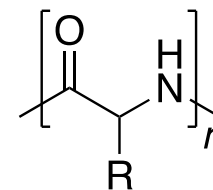
- **Polymers** (“many units”) are very large organic molecules containing a structural motif that repeats many times
- Small molecules that react repeatedly to form polymers are called **monomers**
- Polymers are invaluable in modern chemistry and material science; **biopolymers** are vital in living systems!



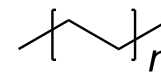
styrene butadiene rubber



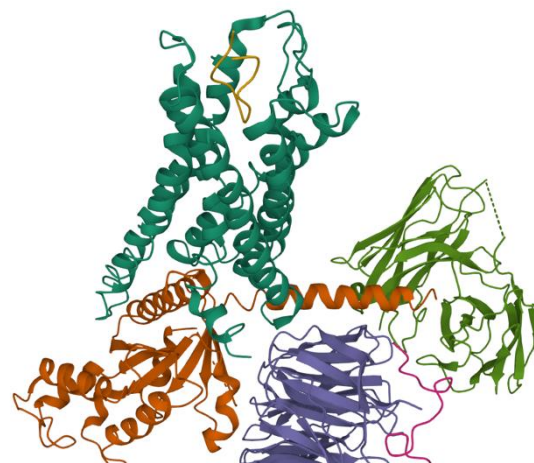
polyacrylonitrile



polypeptide



polyethylene



27.1 Polymers have Tunable and Highly Variable Properties

- Depending on their macromolecular structure, the properties of polymers vary over a wide range. This has led to applications in a huge variety of industries
- **Tensile strength.** From fiber-reinforced plastics to polyurethane foams
- **Conductivity.** From insulating polyethylene to conductive polyacetylene
- **Melting point.** Extremely wide range, affecting robustness and processing costs

Packaging

Electronics

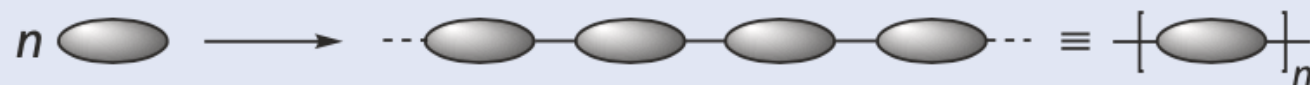
...and so many more!

Automobiles

Construction

27.1 Formation of Polymers from Monomers

- As a monomer reacts with itself, it first forms **oligomers** and eventually polymers
- Addition of one more repeating unit to a polymer does not change the physicochemical properties of the polymer (more or less...)



- Polymers have several different structural aspects that can be tuned to modify their chemistry and properties

Monomer structure(s)

Nature of the bond linking repeating units

Degree of polymerization (n)

Chain architecture (linear, branched, cross-linked)

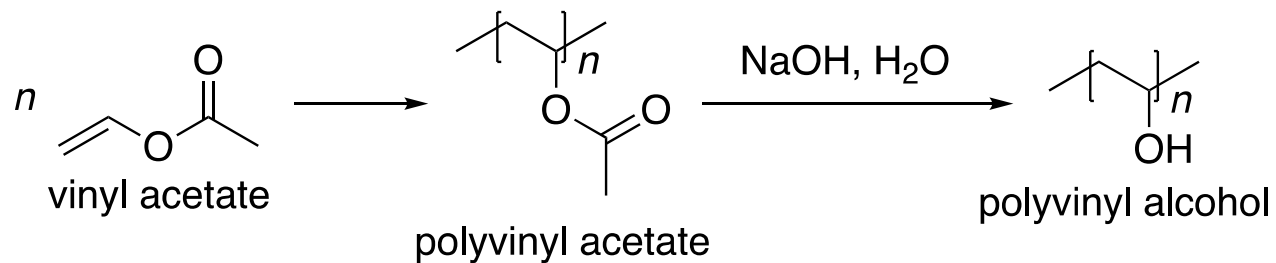
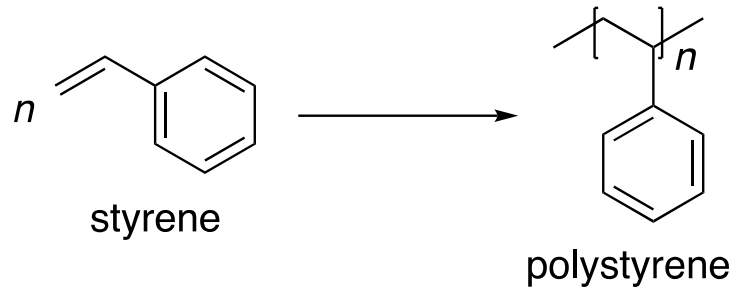
- Polymers have several different structural aspects that can be tuned to modify their chemistry and properties

Arrangement of monomers in a **copolymer**

Noncovalent interactions between chains

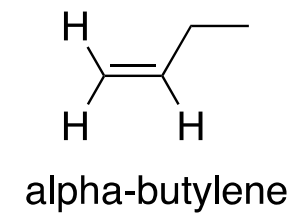
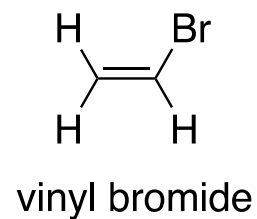
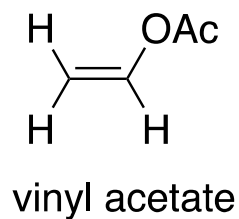
27.2 Nomenclature of Polymers | 1 of 3

- Polymers are typically named using the convention “poly(monomer)”
- Parentheses are often omitted for homopolymers consisting of a single type of monomer



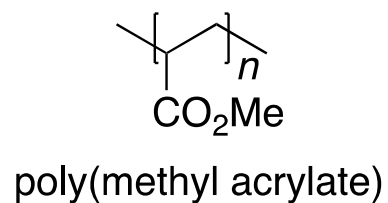
- Polymers are typically named using the convention “poly(monomer)”
- Parentheses are often omitted for homopolymers consisting of a single type of monomer

Example. Draw and name the polymer that results when each of the following monomers undergoes polymerization.



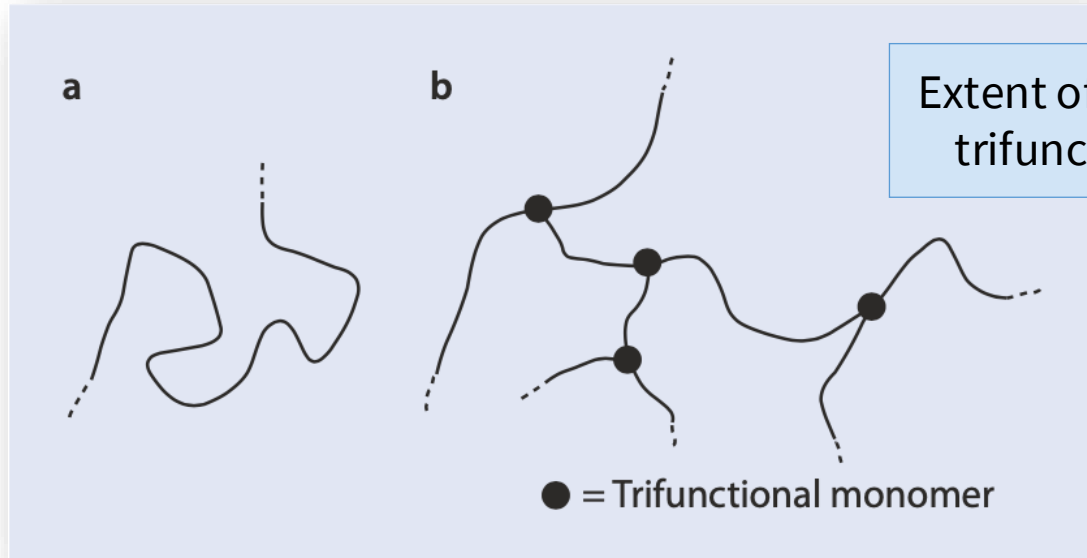
- Polymers are typically named using the convention “poly(monomer)”
- Parentheses are often omitted for homopolymers consisting of a single type of monomer

Example. Draw and name the monomer used to prepare poly(methyl acrylate).



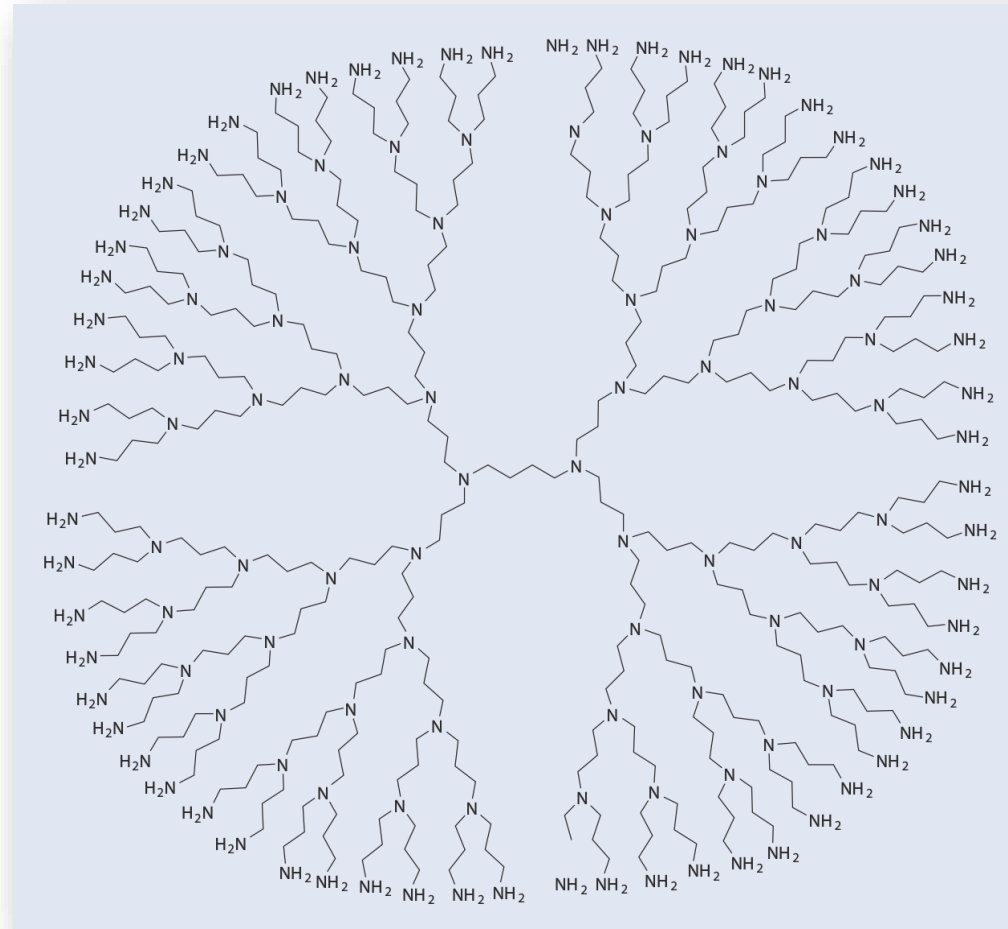
- The introduction of some trifunctional monomers leads to branching points in the polymer structure and **branched polymers**

Linear polymers typically adopt ball-like conformations due to conformational effects.



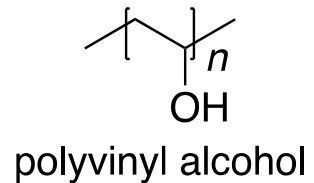
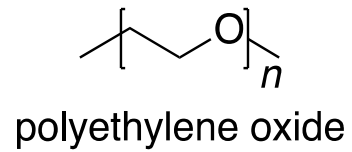
Extent of branching depends on the ratio of trifunctional to difunctional monomers.

- If all monomers are branching points, the result is a maximally branched macromolecule called a [dendrimer](#)
- Dendrimers must be made stepwise and can thus be difficult to synthesize

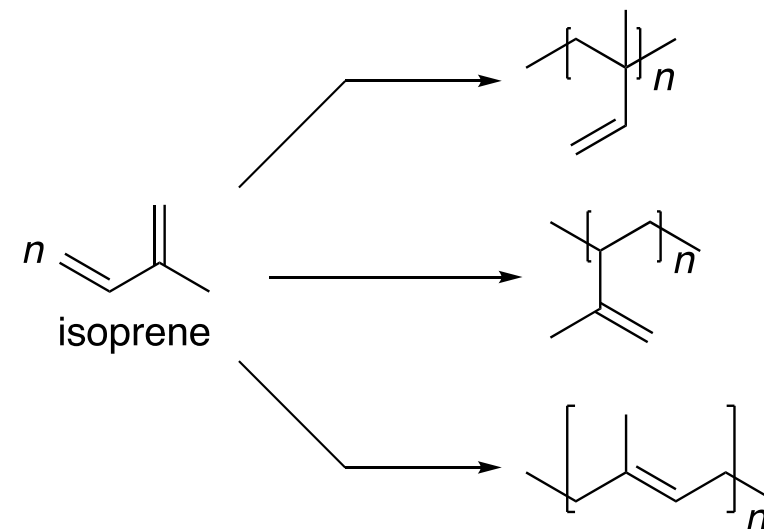


- **Structurally** (or **constitutionally**) **isomeric** polymers have the same atoms in each monomer unit but different covalent linkages

PEO and PVA are structural isomers.



Structural isomers of polyisoprene

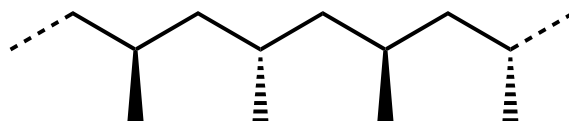


- **Stereoisomeric** polymers have the same connectivity but differ in the spatial positions of groups and the configurations at stereocenters

Isotactic. Identical configurations



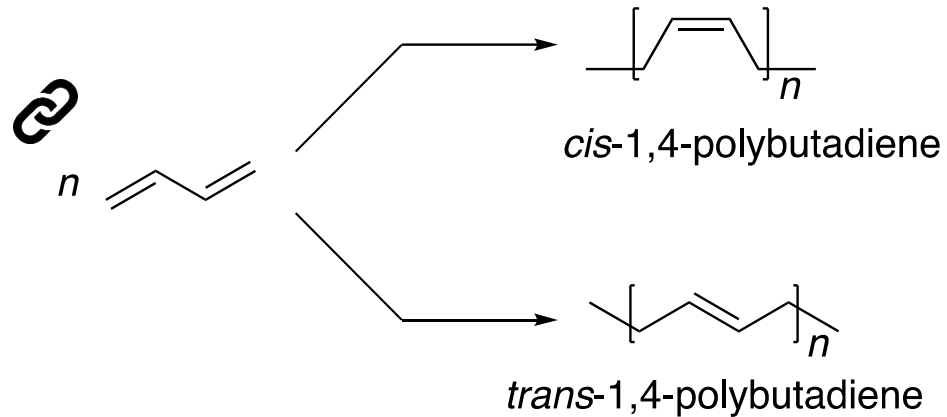
Syndiotactic. Alternating configurations



Atactic. Random configurations



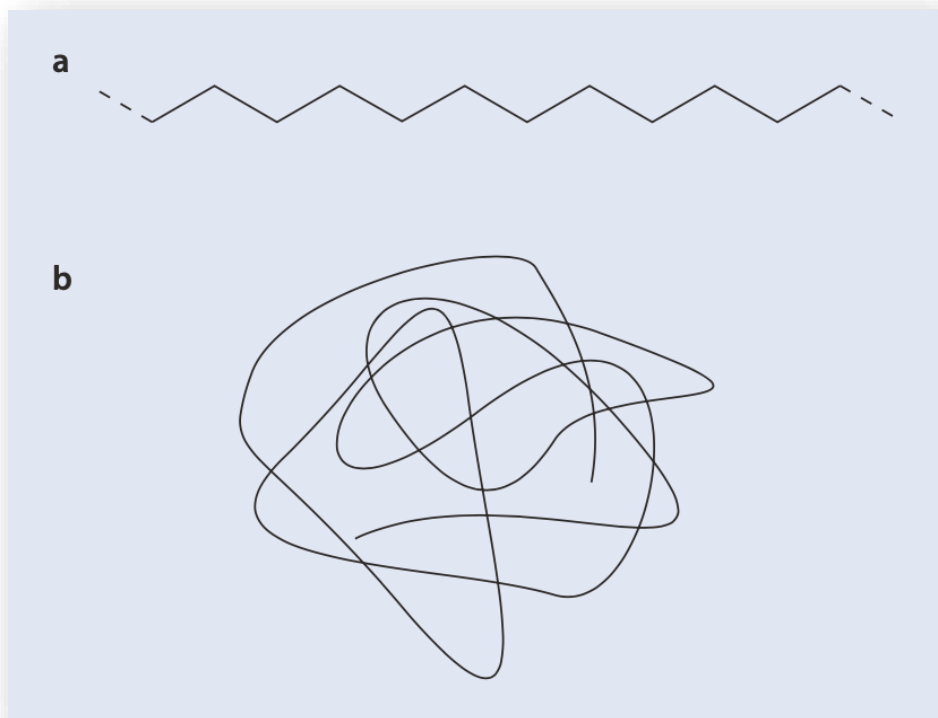
- **Stereoisomeric** polymers have the same connectivity but differ in the spatial positions of groups and the configurations at stereocenters



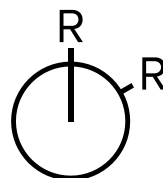
Amorphous, sticky, low melting point

Partially crystalline, high melting point

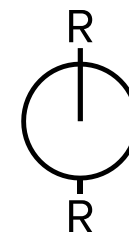
- **Conformationally isomeric** polymers differ in shape due to rotation around bonds in the polymer backbone and/or side chains
- The dominant conformation is usually “spaghetti” or “ball-like”



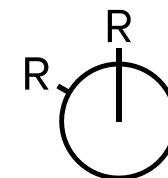
← Enthalpically favorable but entropically unfavorable



0 kJ mol⁻¹
greater *S*



-4 kJ mol⁻¹
lower *S*

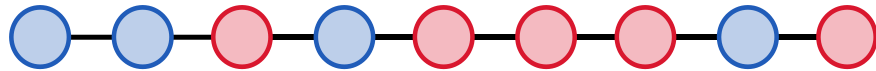


0 kJ mol⁻¹
greater *S*

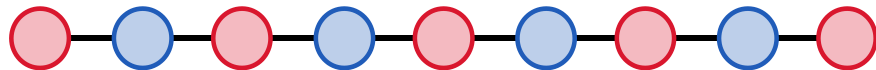
The *anti* : *gauche* ratio is about 1.6:1 in a typical polymer at room temperature.

- **Copolymers** are composed of at least two different monomers
- There are four general ways that comonomers can be arranged in a copolymer

Statistical copolymers contain a (sort of) random arrangement of monomers dictated by relative reactivity and statistics.

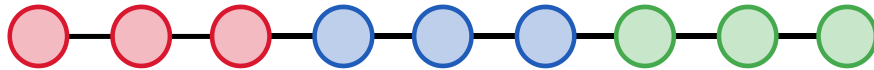


Alternating copolymers contain systematically alternating monomer units.

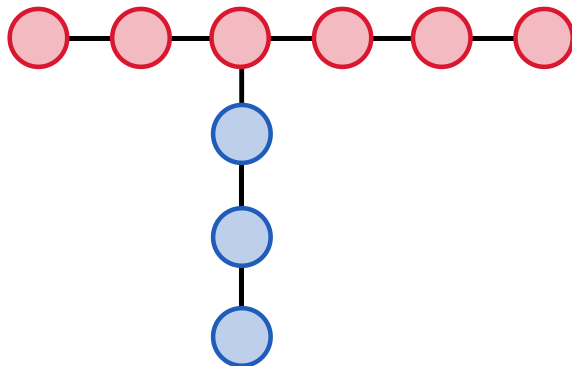


- **Copolymers** are composed of at least two different monomers
- There are four general ways that comonomers can be arranged in a copolymer

Block copolymers contain groups of identical monomer units (blocks). There may be more than just two types of blocks!



Graft copolymers contain distinct monomer units linked to the side chains of a polymer backbone.



- **Copolymers** are composed of at least two different monomers
- There are four general ways that comonomers can be arranged in a copolymer

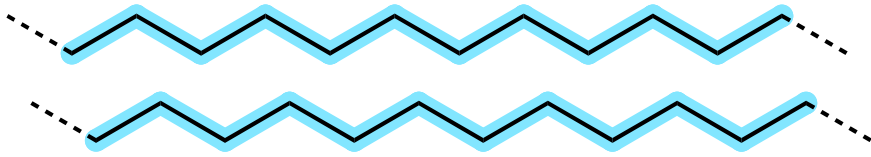
Example. Draw a region of an alternating copolymer constructed from styrene and ethylene.

- **Copolymers** are composed of at least two different monomers
- There are four general ways that comonomers can be arranged in a copolymer

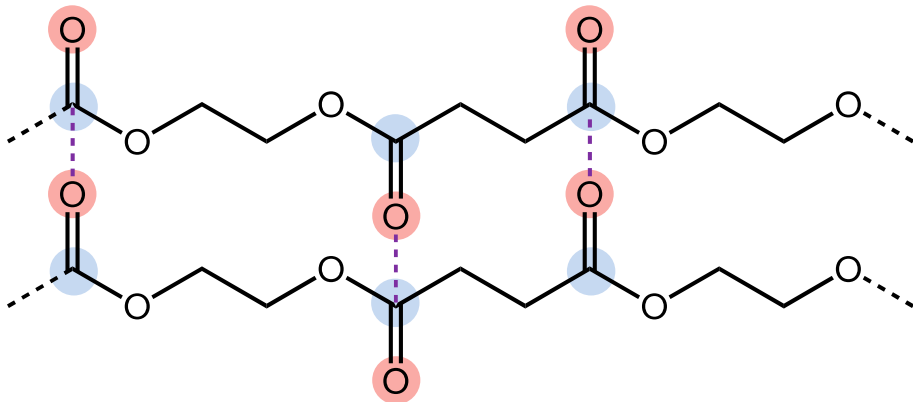
Example. Draw a region of a block copolymer constructed from propylene and vinyl chloride.

- Polymers are subject to the same types of intermolecular forces as small molecules, but often over much larger scales. As a result, emergent effects of small-seeming forces can appear

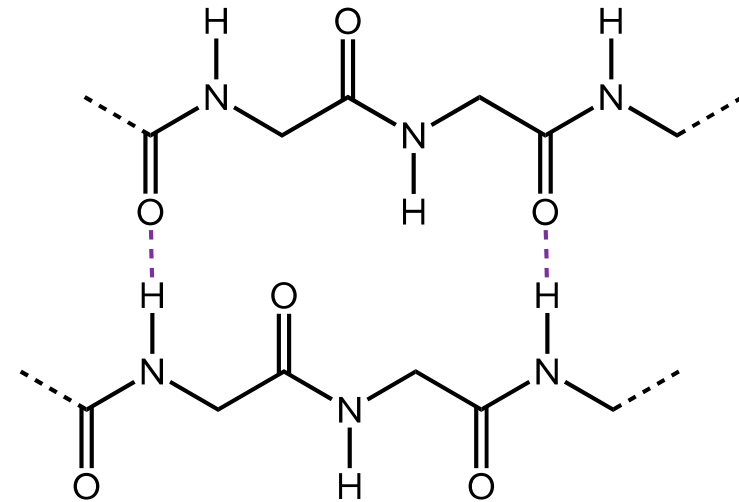
London dispersion forces



Dipole-dipole interactions



Hydrogen bonding



Polymers are Characterized by Molecular Weight Distribution



- Macroscopic samples exhibit a range of molecular weights due to differences in **degree of polymerization (DP)**
- The average molecular weight can be determined in different ways

Number average molecular weight

$$\bar{M}_n = \frac{\sum_i N_i M_i}{\sum_i N_i}$$

Mass average molecular weight

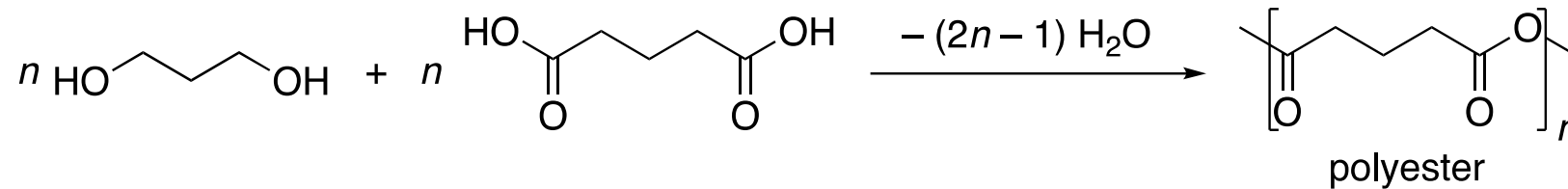
$$\bar{M}_w = \frac{\sum_i N_i M_i^2}{\sum_i N_i M_i}$$

Number average degree of polymerization

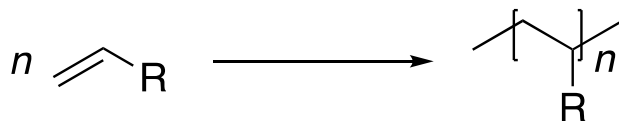
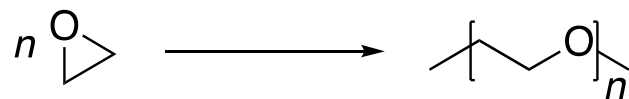
$$\bar{DP}_n = \frac{\bar{M}_w}{M_0}$$

NOTE: Most biopolymers are uniform in length.

- Polymers may be synthesized using **step-growth** or **chain-growth** reactions
- Step-growth reactions are often substitutions or condensations, while chain-growth reactions are often additions
- In a **step-growth** reaction, each monomer bears two reactive functional groups that link together



- Polymers may be synthesized using **step-growth** or **chain-growth** reactions
- Step-growth reactions are often substitutions or condensations, while chain-growth reactions are often additions
- In a **chain-growth reaction**, the growing chain has a reactive functional group that adds to monomer



Chain-growth reactions may involve:

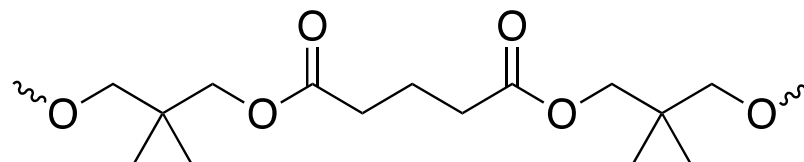
- Radicals
- Anions
- Cations
- Transition-metal catalysis

- Polymers may be synthesized using **step-growth** or **chain-growth** reactions
- Step-growth reactions are often substitutions or condensations, while chain-growth reactions are often additions

Example. Draw the polymer that would be generated from the acid-catalyzed reaction between oxalic acid and resorcinol (1,3-dihydroxybenzene).

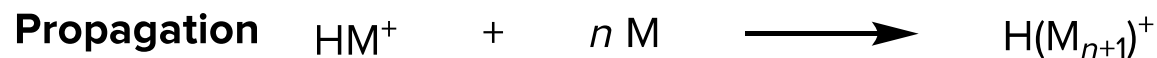
- Polymers may be synthesized using **step-growth** or **chain-growth** reactions
- Step-growth reactions are often substitutions or condensations, while chain-growth reactions are often additions

Example. Determine the monomers you would use to prepare the following condensation polymer.

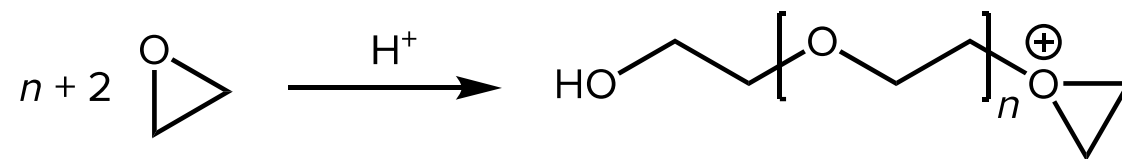


- **Cationic polymerizations** are polyaddition reactions in which the reactive chain end contains a cation
- These reactions are typically initiated by a Brønsted or Lewis acid

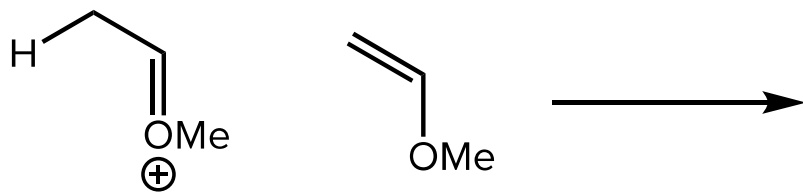
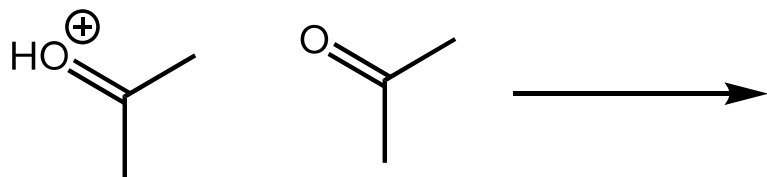
Two-stage mechanism



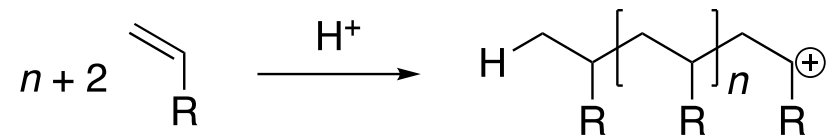
Cyclic, heteroatom-containing monomers



Polymerization of unsaturated groups



- **Nucleophilic** alkenes, in which an electron-donating group is linked to one of the alkene carbons, are ideal for cationic polymerization due to their cation-stabilizing properties



- Anionic polymerizations are polyaddition reactions in which the reactive chain end contains an anion
- These reactions are typically initiated by a Brønsted or Lewis base

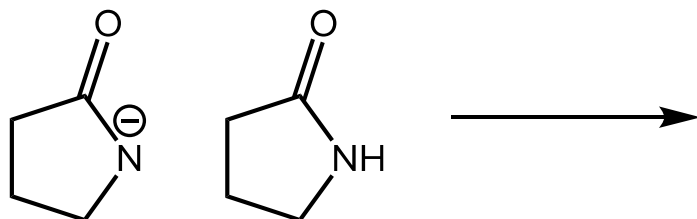
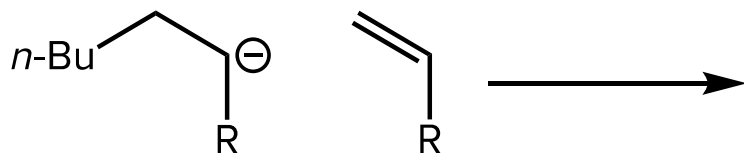
Two-stage mechanism



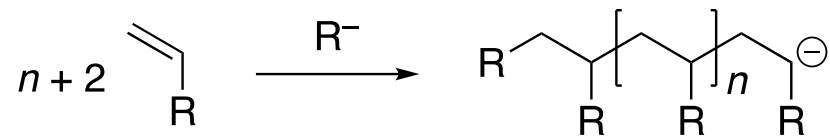
Cyclic, heteroatom-containing monomers



Polymerization of unsaturated groups



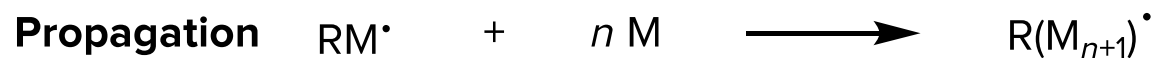
- **Electrophilic** alkenes, in which an electron-donating group is linked to one of the alkene carbons, are ideal for cationic polymerization due to their cation-stabilizing properties



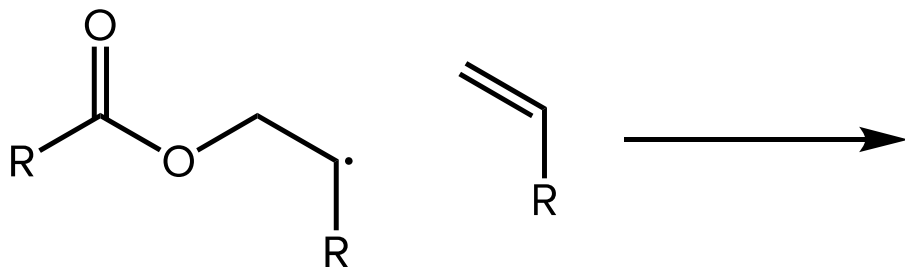
10.11 Radical Polymerization

- **Radical polymerizations** are polyaddition reactions in which the reactive chain end contains a radical
- These reactions are typically initiated by radical initiators containing weak bonds.

Three-stage mechanism



Polymerization of unsaturated groups



Initiators

